

P. Shubham Parashar, Ph.D.

Theoretical Condensed Matter Physics | Quantum Transport and Kinetic Theory

pathaksparashar@gmail.com · psparash@ucsd.edu · Research website · Google Scholar · GitHub · ORCID

Research Profile

Theoretical condensed-matter physicist developing analytical and computational descriptions of transport in spatially inhomogeneous two-dimensional electron systems. My work connects semiclassical Boltzmann kinetics, electron hydrodynamics, magnetotransport, quantum Hall response, and finite-resolution inverse problems. Recent projects address higher-harmonic corrections to hydrodynamic closure, observability in scanning-probe transport, and stochastic worldline mechanisms for heavy-tailed trajectory geometry.

Education and Research Experience

- 2022–2026 **Ph.D., Physics**, University of California San Diego. Advisor: Michael M. Fogler. Dissertation: *Electric and Magnetic Response near Perturbations in Two-Dimensional Electron Gases*.
- 2017–2021 Research with Benjamin Grinstein and Congjun Wu, UC San Diego, on non-associative operators and twist operators on S^7 backgrounds with G_2 instantons.
- 2016–2017 **M.S., Physics**, Indian Institute of Science, Bangalore. Advisor: Rahul Pandit.
- 2012–2016 **B.S., Physics**, Indian Institute of Science, Bangalore. Advisor: Rahul Pandit.
- 2015–2016 Research Intern, Japan Advanced Institute of Science and Technology. Advisor: Ryo Maezono; MEXT-supported.

Publications and Manuscripts

Peer-reviewed

Diffusive and hydrodynamic magnetotransport around a density perturbation in a two-dimensional electron gas

P. Shubham Parashar and Michael M. Fogler

Physical Review B 113, 155443 (2026). arXiv:2601.16402.

Nanoscale imaging of magnetotransport around a circular p–n junction in graphene

Z. J. Krebs, W. A. Behn, K. J. Smith, M. A. Fortman, K. Watanabe, T. Taniguchi, **P. Shubham Parashar**, M. M. Fogler, and V. W. Brar

Physical Review Letters 136, 136301 (2026); Editors' Suggestion. arXiv:2409.19468.

Under review / submitted / in preparation

Fourth-order closure obstruction and chiral nonlocality in circular kinetic magnetotransport

P. Shubham Parashar

Under review at *Physical Review B* (2026). arXiv:2607.22730.

Thermal and viscous contrast in quantum Hall scanning-probe images

P. Shubham Parashar

Submitted to *Physica E: Low-dimensional Systems and Nanostructures* (2026). arXiv:2607.16315.

Diagnosing changes in trajectory-geometry tails: endpoint laws, crossovers, and finite-window tests

P. Shubham Parashar

Preparing for submission to *Physica A* (2026).

Selected Contributions

- Derived magnetotransport around smooth density perturbations across Ohmic and hydrodynamic regimes, including Landauer resistivity dipoles and Gurzhi-scale redistribution.
- Identified the fourth-order current operator omitted by stress-level hydrodynamic closure and tracked its chiral finite-wave-number completion into the nonlocal regime.
- Formulated finite-resolution observability tests for thermoelectric and Hall-viscous response in quantum Hall scanning-probe measurements.

Awards, Methods, and Selected Presentations

Awards: Bhadra Fellowship (2024); NSF-supported graduate research (2018–2020); UC San Diego Scholar's Grant (2017); KVPY Fellowship (2012–2017); MEXT-supported research internship (2015–2016).

Methods: Boltzmann-harmonic hierarchies, hydrodynamic and higher-gradient closures, Green-function and boundary-value methods, asymptotic and Mellin analysis, stochastic path integrals, inverse problems, convergence and uncertainty tests.

Tools: Python, MATLAB, Mathematica/Wolfram Language, C++, Fortran, $\text{\LaTeX}$, Git.

Selected talks: Ph.D. dissertation defense, UC San Diego (2026); APS March Meeting (2020, 2018); Columbia University (2022).